



SCHOOL OF ENGINEERING AND TECHNOLOGY BULLETIN

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**WITH THE BLESSINGS OF REVERED
PROF. DR.VISHWANATH D. KARAD**



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UNDER THE GUIDANCE OF



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WORLD PEACE PRAYER

ॐ नमोजी जाद्या । वेद प्रतिपाद्या । जयजय स्वयंवेद्या । आत्मरूपा ।।१।।
देवा तुंचि गणेशु । सकलार्थ मतिप्रकाशु । म्हणे निवृत्तिदासु ।
अवधारितो जी ।।२।।

गुरुब्रह्मा गुरुर्विष्णुः गुरुर्देवो महेश्वरः । गुरुः साक्षात् परब्रह्म तस्मै श्री
गुरवेनमः ।।

ॐ पूर्णमदः पूर्णमिदं पूर्णात् पूर्णमुदच्यते । पूर्णस्य पूर्णमादाय
पूर्णमेवावशिष्टे
।। ॐॐॐॐ शान्तिः । शान्तिः । शान्तिः ।।

हरि ॐ ईशा वास्यमिदं सर्वम् । यत्किंच जगत्यां जगत् ।। तेन त्यक्तेन
मुंजीथाः । मा गृधः कस्यस्विद् धनम् ।।

ॐ भूर्भुवः स्वः । तत्सवितुर्वरेण्यं ।। भर्गोदेवस्य धीमहि । धियो यो नः
प्रचोदयात् ।।

सर्वेऽपि सुखिनः सन्तु । सर्वेऽसन्तु निरामयः सर्वेऽभद्राणि पश्यन्तु । मा
कञ्चिद् दुःखमाप्नुयात्

द्यौः शान्तिः । अन्तरिक्षं शान्तिः । पृथ्वी शान्तिः । आपः शान्तिः ।
औषधयः शान्तिः । वनस्पतयः शान्तिः । विश्वेदेवाः शान्तिः । ब्रह्म
शान्तिः । सर्व शान्तिः । शान्तिरेव शान्तिः । साऽमा शान्तिरेधि ।।१७।।
ॐॐॐ शान्तिः । शान्तिः । शान्तिः ।

Department Initiatives & Strategic Developments

The Department of Civil Engineering, MIT-WPU achieved a significant milestone by successfully organizing the International Workshop on "Sustainable Tunneling for a Better Life" on 23rd–24th June 2025. The workshop brought together distinguished international experts and industry professionals to share advanced knowledge on sustainable tunneling and infrastructure development. The inauguration by Shri Nitin Gadkari, Hon'ble Minister of Road Transport and Highways, added great prestige to the event. This achievement highlights the department's commitment to academic excellence, global collaboration, and contemporary infrastructure practices.



Practice-Driven Conceptual Excellence- A New Pedagogy for FY was implemented which helped in greater understanding of the courses.

By the student for the students assignments were conducted for some of the courses. Mapping of CCA with performance of competitive coding platform for the courses-DSA & OOP.

Value added course on Cloud for Beginners – Understanding of Cloud and Ecosystem was conducted by International Faculty.

Triangle Trophy, APAS, and CHAP event was conducted.

CSR-SLB funds of Rs. 7 Lakhs was received.

Hardware and software licences of Rs. 121 Lakhs were received as sponsorship under Robocon.

Corporate training worth Rs.34.5K was conducted under eMobility Centre of Excellence.

Guest Lecture Report on Forging Processes & Design Considerations

Date: 9th April 2025

Speaker: Mr. Ranjit Dalwatkar, Design Engineer, Regal Rexnord, Pune



Development of Centre of Excellence for Hydrogen Generation and Storage led by Dr. Sagar P. Kanekar, Dr. Santosh Patil, Dr. Anand D. Kulkarni, Dr. Dev Kumar Thapa, Dr. Soumava Biswas, Dr. Anand J. Kulkarni, Dr. Rajib Sinharay, Dr. Sharbani Kaushik, Ms. Sonali Saxena



Technophilia 2025 was a two-day state-level event conducted by MIT Polytechnic and School of Polytechnic and Skill Development, Pune. The event featured a mix of technical and non-technical competitions designed to test and showcase the knowledge, creativity, problem-solving ability, and practical skills of diploma-level students from various institutions. The primary objective was to foster innovation, team spirit, and inter-collegiate collaboration through well-structured, theme-based events.

Technophilia 2025	
Technical Events	Non-Technical Events
Bugzapper Model Making Welding Competition Kicad Challenge	The Art of Vision Slow Cycling

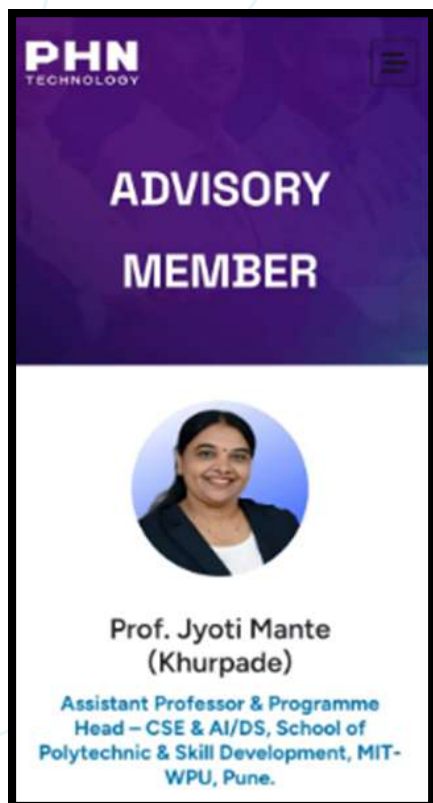


Expert Lectures



An expert lecture on topic "Analog VLSI Design and Fabrication" was conducted on 17th March 2025, delivered by Dr. Mosami Bhanja, Assistant Professor, Symbiosis Institute of Technology.

Faculty Achievements & Recognition



Prof. Jyoti Mante(Khurpade), Programme Head,Computer Science and Engineering (CSE) and Artificial Intelligence and Data Science (AIDS), Department of Polytechnic,MITWPU has been appointed as an Academic Advisory Member on the Board of PHN Technology Pvt. Ltd. This prestigious recognition highlights Prof. Jyoti's exceptional expertise and leadership in the field of computer science and emerging technologies. Her role on the board will provide valuable strategic guidance and insights, further strengthening the collaboration between academia and industry.

Prof Christian Preinfalk from Switzerland visited Bioengineering lab and met faculty & students working on biohydrogen production. The visit was fruitful in terms of collaboration for scale up production.



Department of Electrical and Electronics Engineering has signed the MoU with Teckschool skill hub LLP on 26th June 2025. The MoU will facilitate students of M. Tech. (VLSI) will get hands on training on tools like Synopsys and strengthen industry internship and placement.



Department of Electrical and Electronics Engineering has signed the MoU with TM2Space Technologies Pvt Ltd 14th May 2025 for technical guidance for STeRG projects.

Department of Electrical and Electronics Engineering has signed the MoU with Waree Energies Limited on 29th April 2025. This MoU aims to strengthen students industry internship and Placement, conduct Guest lecture, and undertake Research project in the field of Solar energy.

Department of Electrical and Electronics Engineering has signed the MoU with M. A. Rangoonwala College of Dental Sciences & Research Centre, Pune on 14th April 2025 for consultancy projects and collaborative research projects in the field of dentistry and engineering.



Department Initiatives – MOUs

1. Laser Technologies Pvt. Ltd., Raigad, Maharashtra

An Department of Mechanical engineering has signed MoU with Laser Technologies Pvt. Ltd., Raigad, Maharashtra to promote collaboration between the institution and the company. This partnership aims to support knowledge exchange, industry interaction, and practical exposure for students through joint activities and initiatives.



2. Aluminium Casters' Associations (ALUCAST), Baner, Pune, Maharashtra

The institution also signed an MoU with Aluminium Casters' Associations (ALUCAST), Baner, Pune, Maharashtra. This collaboration focuses on strengthening industry–academic relationships, encouraging technical cooperation, and providing opportunities for skill development and industry engagement.

3. Bombay Industries Association, Kurla Industries Estate, Mumbai, Maharashtra

Another MoU was established with the Bombay Industries Association, Kurla Industries Estate, Mumbai, Maharashtra. The agreement is intended to enhance industry exposure, facilitate knowledge sharing, and create platforms for collaborative programs and professional development.



4. Tata Technologies Ltd., Rajiv Gandhi Infotech Park, Hinjewadi, Pune

The institution signed an MoU with Tata Technologies Ltd., Rajiv Gandhi Infotech Park, Hinjewadi, Pune to strengthen ties with the technology industry. This partnership aims to support innovation, industry training, and collaborative initiatives that benefit students and faculty.

Dr. Anuradha Phadke, Associate Professor in Department of Electrical and Electronics Engineering was elected as Ordinary Member of the Institution of Engineering and Technology (IET) Pune local network executive committee for the year of 2025.

Dr. Shweta Kukade, Dr. Jyoti Lele and team from Department of Electrical and Electronics Engineering won 1st prize in Ideathon held in March 2025 (HackWPU25).

Team IOTIDE received two awards– Ideathon Runner up and Wirkathon Appreciation award under the mentorship of Dr. Anuradha Phadke in HACKMITWPU, Hackathon organized by MITWPU held on 26 to 28 March 2025. The team members are Soham Walimbe, Chaitanya Abhyankar, Revati Jagdale, Anurag Sharma, Aayush Deshmukh and Shriya Samruddhi.



Team “Marathi LLM” won “Best Research based Project Award” in the Texepher 2025 project competition under the mentorship of Dr. Anuradha Phadke held on 16 Apr 2025. Project Team members are Nishant Kshirsagar, Sarthak Patwari and Aman Singh.

Prof. Bhakti Paranjape, Dr. Apurva Naik received the Appreciation award for presenting idea in the PhD track of HackWPU25.



Department of Bioengineering organized BioConclave and invited industry people from all over India to discuss the possibilities of collaboration for internship and placement. A hands on training fermentation workshop was also organized for better understanding of bioprocess engineering elective.



The Department of Civil Engineering has impactful collaborations with national and international institutions to enhance academic quality, industry engagement, and global visibility.

The organization has agreed in principle to set up a Tunneling Jumbo simulator at the college, subject to resolution of tax-related challenges, with an approximate contribution value of ₹90 lakhs. Additionally, miniature models of a Tunneling Jumbo and a Loader (₹45,000 each) will be provided this week.

This Memorandum of Understanding (MoU) outlines a collaborative agreement between Dr. Vishwanath Karad MIT World Peace University and Tejus Infratech LLP for academic–industry cooperation. The partnership aims to strengthen technical education, training, and practical exposure in civil and infrastructure engineering.

MIT World Peace University (MIT-WPU), Pune has developed strong academic, industry, and government collaborations to enhance education, research, and practical learning in civil and infrastructure engineering. A lifetime Memorandum of Understanding with the Indian Concrete Institute (ICI) supports research activities, student internships, and sustained industrial exposure, as demonstrated by the successful organization of the national-level CEMCON 2025 conference. This long-term association strengthens interaction between academia, researchers, and industry professionals.

Further extending its national engagement, MIT-WPU has signed an MoU with the Ministry of Road Transport and Highways (MoRTH), Government of India, to promote education, training, and research in transportation and highway engineering. This collaboration aligns academic programs with national infrastructure priorities and provides students with exposure to real-world projects and policy-driven practices.

In addition, an industry partner has agreed in principle to establish a Tunnelling Jumbo Simulator at the university, with an approximate contribution value of ₹90 lakhs, subject to resolution of tax-related challenges. To support immediate learning, miniature models of a Tunnelling Jumbo and a Loader are also being provided. The university has also entered into an MoU with Tejus Infratech LLP to strengthen academic–industry cooperation through training, internships, expert interactions, and practical exposure. These initiatives collectively enhance experiential learning and industry readiness of students.



The visit of Head,Cognizant to Subsea lab accompanied my Dean,SOET and Dr Siraj Bhatkar



The industrial visit to Shell Technologies, Quest Global and Chevron, Bangalore by Dr Siraj Bhatkar and Dr Vinayak Wadgaonkar

Industrial Visit

Industry visits provide students with valuable hands-on experience and insights into real-world applications of their academic knowledge. These visits allow students to observe cutting-edge technologies, interact with industry experts, and understand the practical challenges and solutions in various sectors

- A total of 70 students of SYCSE visited the IT Department of Bank of Maharashtra, Pune, on 17th April 2025 as part of an academic visit.



- A group of 40 students from SYAIDS visited TomTom India Pvt. Ltd., Yerwada, on 12th April 2025 as part of an educational and industrial visit.



A total of 70 students from SYCSE–B visited TechnoGrowth Software Solutions Pvt. Ltd. on 12th April 2025 as part of an industrial visit. The visit provided students with exposure to real-world software development practices. They gained insights into industry workflows, emerging technologies, and professional standards. The interaction helped bridge the gap between academic learning and practical application.



A group of 40 students from SYAIDS visited Tron SofTech Pvt. Ltd. on 5th April 2025 as part of an industrial exposure program.

The Electrical and Computer Engineering students along with Dr. Netra Lokhande and Prof. Manisha Wani of Department of Electrical and Electronics of MIT WPU visited the 220kV Parvati Substation in Pune on 28th March 2025. The Substation is operated by MAHATRANSCO, and is a crucial transmission substation handling 220kV, 132kV, and 22kV power levels. The Substation serves as a key grid node for Pune's power supply, featuring transformers (50, 100, 200 MVA), and SCADA systems.

Alumni Guest Lecture Mr. Arpit Bendale, General Manager, APL Logistics (I) Ltd, Pune.



A site visit to the Sewage Treatment Plant (STP), Erandwane, Pune, was conducted for the SY B Tech students to provide hands-on exposure to wastewater treatment processes. Participants observed various treatment units, including preliminary, primary, secondary, and tertiary treatment systems. The visit enhanced understanding of process flow, operational control, and regulatory compliance in sewage management. It also highlighted the importance of sustainable wastewater treatment and reuse practices. Overall, the visit strengthened practical knowledge and reinforced classroom concepts with real-time applications.



The industrial visit to Worldwide Oilfield Machine Bibwewadi, Pune for Third Year and Final Year UG student (Petroleum Engineering) along with Dr Kunal Modak

Faculty Research Articles

Department of Civil Engineering:



(Dr. Pravin Minde)

Offsite construction, using prefabrication and modular methods, is gaining traction in India to tackle delays, cost overruns, and labour shortages. This approach can reduce project timelines by 30–50%, enhance quality control, and minimize waste, making it a sustainable solution for rapid urbanization (Dr. Pravin Minde).



(Dr. Adinath Yadav)

Improvement of torsional rigidity in steel supporting structure topside modules enhances global stiffness and reduces excessive twist under combined wind, wave, and operational loads. Increased torsional rigidity improves load redistribution among members, leading to better structural stability and fatigue performance. It also minimizes differential deformations that can adversely affect equipment alignment and serviceability. Research in this area supports optimized bracing configurations and material utilization for safer and more efficient offshore topside designs. (Dr. Adinath Yadav)



Mrs.M.P.Fatangare
Department of Polytechnic

Nadi Pariksha is an ancient diagnostic process in Ayurveda in which a practitioner assesses the pulse of a patient and identifies imbalances with the doshas— Vata, Pitta, and Kapha— that represent the three energies that govern all bodily and mental functions. The doshas must be in balance for one to be healthy. A practitioner places his or her fingers feeling for the pulse on the radial artery at the wrist and assesses the pulse for its speed, rhythm, and distinct qualities. This process is difficult to master because of how much artistry and scientific observation is involved. Also, the goal of Nadi Pariksha is to identify disease early on before the disease has expressed its full symptoms and to prescribe a personalized plan/treatment to restore a person's physical, emotional, and mental health. Nadi Pariksha, based on the subtle vibratory patterns that can be palpated, gives the practitioner detailed information on every aspect of health linked to organ functions and physiological states.

The emergence of artificial intelligence (AI) is now bridging the gap between ancient wisdom and modern diagnostic practices. The digital format of Ayurvedic Nadi Pariksha makes use of machine learning algorithms that analyze pulse waves that are obtained by electronic sensors that are placed into a watch and arranged on the wrist. This interface denies the emphasis of subjective interpretation and looks to enhance objectivity in any situation by analyzing the pulse's characteristics using science by demonstrating a numerical presentation of discrepancies and relating these to dosha imbalances and health problems. AI systems can analyze enormous amounts of data and identify patterns and variations in the pulse that the human brain may not be able to comprehend. This melding of traditional medicine with contemporary

The association of Ayurvedic Nadi Pariksha with AI-powered Nadi Pariksha follows a synergistic relationship between traditional Ayurvedic diagnostics, which implements an ancient holistic practice, with modern-day state-of-the-art technologies. This transition represents a unique opportunity to transform our current definition of health care and place focus on prevention and individualized medicine.

Research Publications:

Journals :

Sr. No.	Authors	Title	Year	Source Title	Volume & Issue	Page No / Article No
1	J. Mante; K. Kolhe	A novel DDoS detection framework leveraging hybrid binary grey wolf optimization and explainable AI	2025	Journal of Information & Optimization Sciences	Vol. 46, Issue 2	563–578
2	A. Bakare; N. Patil; D. Shastri	Review of Diffuser Augmented Wind Turbines and development of it in Urban areas	2025	Journal of Scientific Advances	Vol. 2, Issue 01	13–30
3	A. S. Chougale; S. S. Wagh; A. D. Waghmare; S. R. Jadkar; D. R. Shinde; S. F. Shaikh; H. M. Patil	Tailoring ZnO–CdO Nanocomposite Photoanodes Mimicking Neural Web Like Structure for Optimized Solar Water Splitting	2025	Advanced Energy and Sustainability Research	—	Article No. 2500009
4	D. S. Shastri; M. R. Mahajan; A. Sur; V. W. Bhatkar	Electric Vehicle Battery Cooling Using Phase Change Materials and Nanofluids	2025	International Journal of Heat & Technology	Vol. 43, Issue 3	—

Research Publications:

Sr. No.	Authors	Title	Year	Source Title	Volume & Issue	Page No / Article No
1	N. Pradhan; B. S. Chaudhari; P. D. Khandekar	Adaptive Slotframe Allocation with QoS and Energy Optimization in 6TiSCH for Industrial IoT Applications	2025	<i>Telecom</i>	Vol. 6, Issue 2	Article No. 41

Journals:

Sr. No.	Authors	Title	Year	Source Title
1	G. D. Shrigandhi; B. S. Kothavale	Hoop tensile strength testing of glass/jute hybrid filament wound composite tubes	2025	International Journal on Interactive Design and Manufacturing
2	A. Ahuja; D. Waghole; S. S. Ramdasi	Fuel consumption optimization in solid oxide fuel cell based hybrid electric tractor using multihead cross attention and sea-horse optimization	2025	Optimal Control Applications and Methods
3	P. K. Tamkhade; V. A. Naikwadi; K. G. Kulkarni; R. N. Todkar; H. C. Pisal; M. D. Joshi	Design, development and analysis of seat constrainer system for a car	2025	International Journal of Vehicle Structures and Systems
4	P. M. Patane; S. S. Patil; M. R. Nandgaonkar	Investigation of combustion characterization of single-point and two-point laser ignition of LPG-air mixture in a constant volume combustion chamber	2025	Combustion and Flame
5	T. Bansal; V. Patel; M. Solanki; M. Bumb; K. Deshmukh	Developing a primitive humanoid legged locomotion to gain an understanding of human mobility	2025	Evergreen

6	P. Pesode; S. Barve	Biocompatibility of plasma electrolytic oxidation coated titanium alloy for biomedical applications	2025	BioNanoScience
7	K. V. Raibole; K. V. Mali	Experimental and computational study on a morphed trailing edge airfoil for enhanced lift and stall characteristics	2025	International Journal of Sustainable Aviation
8	R. Shinde; M. N. Pradhan; B. Sahoo; R. Jain	Failure analysis of a torsion spring: A microstructural and finite element assessment	2025	Engineering Failure Analysis
9	R. C. Nayak; M. K. Roul; M. V. Kulkarni	Design of an innovative fuel-efficient stove for reducing air pollution and enhancing sustainability	2025	Environmental Progress and Sustainable Energy
10	N. Anekar; P. Selmokar; N. Satpute	Performance analysis of lateral sliding mode triboelectric nanogenerators for self-powered healthcare applications	2025	Journal of Electrostatics
11	A. Srivastava; P. Meena; P. Patane; D. Meena; S. Shelare; C. Wagle	Advancements in biogas digester materials: A review of strength, durability, and suitability	2025	Water Environment Research
12	P. K. Meena; D. Kumar; S. Shelare; S. Kumar; P. M. Patane; C. S. Wagle; S. D. Awale	Factors influencing biohydrogen production from waste: Insights into substrates and operating conditions	2025	Environmental Progress and Sustainable Energy
13	A. Srivastava; P. K. Meena; P. M. Patane; D. Meena; S. Shelare; C. S.	Sustainable biogas purification techniques using indigenous materials: A review focused on developing economies	2025	Biofuels

14	S. K. Pandey; D. Kumar; S. Dwivedi; A. R. Dixit	Mechanical strength and shrinkage investigation of fused filament fabricated H13 die steel	2025	Lecture Notes in Mechanical Engineering
15	D. Kumar; S. Dwivedi; A. K. Srivastava; A. R. Dixit	Comprehensive study on mechanical response of strut-based and sheet-based 316L stainless steel cellular architectures for biomedical application through laser powder bed fusion	2025	Materials Today Communications
16	C. S. Kumar; S. Gupta; A. K. Sharma	Characterization of TiN-HA nanocomposite developed through rapid microwave sintering route at high temperature	2025	International Journal of Applied Ceramic Technology

Journals:

Sr. No.	Authors	Title	Year	Source Title
1	P. Minde; J. Patil	A comprehensive review of factors affecting 3D printing technology adaptation in the Indian construction sector	2025	Journal of Building Pathology and Rehabilitation
2	R. R. Salgude; M. Ghag; V. Sakhare; S. Shekokar	Sustainable building design: Optimizing building geometry and external envelope to minimize energy consumption and CO ₂ emissions	2025	Asian Journal of Civil Engineering
3	S. Sawant; R. Salgude; S. Shekokar; S. Dalvi	Analysing the building information modeling for construction in India – A review	2025	Lecture Notes in Civil Engineering (Springer)
4	M. Z. Kangda; A. Ubaid; S. Sathe; V. Wadki; S. W. N. Razvi; R. G. Raikar	Shape memory alloys: A resilient structural engineering material	2025	Multiscale and Multidisciplinary Modeling, Experiments and Design
5	N. S. Pawar; K. V. Sharma	Comprehensive review of remote sensing integration with deep learning in landslide forecasting and future directions	2025	Natural Hazards
6	G. U. Kiran; N. Ganesan; D. Roy; S. N. Shinde; M. Indumathi; G. U. Alaneme; V. H. U. Eze; K. Anton	Evaluation of net-zero materials in mortar bricks with predictive modelling using random forest and gradient boosting techniques	2025	Discover Applied Sciences
7	G. Meena Devi; S. N. Shinde; J. Leo Amalraj; S. Isaac; A. Chouksey; D. Zunaithur Rahman	Impact of marble sludge powder on the mechanical properties of fiber-reinforced concrete: Experimental and ANN prediction	2025	Computers and Concrete
8	S. R. Ahir; S. Pimplikar	Comparative analysis of residential property rates and rents for West Pune City pre and post COVID	2025	AIP Conference Proceedings

9	S. S. Dhodare; N. Gogate; R. Dod	Eco-bricks reinvented: Transforming coal mine overburden and fly ash into sustainable building solutions	2025	Innovative Infrastructure Solutions
10	S. P. Banne; G. S. Ingle; A. Dhawale; S. Gaikwad; S. Bobade	Experimental study on the stabilization of laterite soil with chitosan biopolymer for subgrade applications	2025	Engineering and Applied Science Research
11	R. V. Patil; M. B. Sonawane	Identification and prioritization of risk factors impacting cost overrun in Indian road construction projects	2025	Lecture Notes in Civil Engineering (Springer)
12	B. Achour; L. Amara; K. H. Kulkarni	In-depth investigation of flow measurement using sharp-edged width constriction in trapezoidal open channels	2025	Larhyss Journal
13	A. Yadhav; M. Kulkarni; D. Ingale	Improvement of torsional rigidity of steel supporting structure topside module	2025	Journal of Mines, Metals and Fuels
14	P. Hanjage; A. G. Shelake	An experimental study on the effects of PVA, jute, and Recron 3S fibers on bendable concrete properties	2025	Lecture Notes in Civil Engineering (Springer)
15	S. Dandin; S. Sathe; G. Ingle; R. Rathod	Influence of biopolymers on geotechnical properties and erosion resistance of colliery spoil dumps	2025	Environmental Science and Pollution Research
16	P. Minde; M. Kulkarni; D. Burde	Exploring the quality and safety aspects of LGSF-Ferron modular construction in Indian context	2025	Journal of Mines, Metals and Fuels

17	V. Sakhare; N. Khairnar; U. Dahatonde; S. Mashalkar	Review on sustainability in 3D concrete printing: Focus on waste utilization and life cycle assessment	2025	Asian Journal of Civil Engineering
18	S. Sathe; D. A. Kolte	Enhancing concrete performance with hybrid basalt–polypropylene fibers: Experimental and numerical evaluation	2025	Innovative Infrastructure Solutions
19	S. Sathe; B. Gholap	Influence of aggregate gradation on the performance of fibre-reinforced pervious concrete	2025	Proceedings of the Institution of Civil Engineers: Structures and Buildings
20	S. Sathe; C. U. Pawar	Exploring the effectiveness of polypropylene fiber-reinforced geo-polymer concrete	2025	Journal of Mines, Metals and Fuels
21	P. Kushwaha; L. Gautam; K. V. Sharma; A. S. Chauhan; B. L. Swami	Impact of Sasobit and Zycotherm additives on the performance of warm mix asphalt: Enhancing rutting resistance and moisture durability	2025	Journal of Building Pathology and Rehabilitation
22	H. S. Chauhan; K. V. Sharma; P. K. Gautam	Performance evaluation predictive modeling of cement mortar durability with dimensional limestone waste	2025	Journal of Building Pathology and Rehabilitation
23	L. Mishra; P. N. Bhowmik; P. Anand; S. Yantrapalli	A study on de-contamination of dairy wastewater using natural coagulants	2025	Environmental Monitoring and Assessment

Book chapter

Authors	Title	Year	Source Title	Volume / Type	Page No
M. S. Joshi; V. V. Pangave	Use of NLP Techniques and High-Performance Computing for Automated Knowledge-Based Ontology Construction of Saffron Crop	2025	<i>Parallel and High-Performance Computing in Artificial Intelligence</i> (Auerbach Publications)	Book Chapter	160–180

Patent

Sr. No.	Patent Title	Inventor(s)	Status
1	Rebar binding tool	Prof. Sagar Sonawane; Prof. RatanKumar Patil	Granted

Sr. No.	Patent Title	Inventor(s)	Application No.	Status	Publishing Authority
1	A Modular Unmanned Ground Vehicle with Multi-Modal Sensing and Control Capabilities	Aher Jaideep Vinod; Gadekar Abhijit; Vibha Patel	202424065762	Published on 23 May 2025	Government of India
2	A method for enhancing emotional communication between humans and animals	Shreyas Shinde; Dr. Mrunal Annadate; Dr. Jagdale	202521038338	Published on 09 May 2025	Government of India
3	IoT-Based Web Switch Motor Control for Remote Automation	Dr. Ujjwala Rawandale; Kabeer Latane; Priyanshu Krishna; Sahil Hunswadkar; Dr. Trushita Chaware	202521040737	Published on 30 May 2025	Government of India

Sr. No.	Patent Title	Inventor(s)	Application No.	Status	Country
1	A Water Cleaning Boat	Patel Hitik; Kulkarni Kaustubh Ganesh; Tamhankar Neel; Havaladar Sanjay Narayan	202424071843	Published	India
2	A Soft Underwater Multiarmed Robotic Gripper	Waghmare Pratap; Thote Abhishek M	202424071842	Published	India
3	A Levitating Device for Vehicles	Chaitanya Mandar Sahasrabudhe; Daman Chakraborty; Purnesh Jain; Dr. Omkar K. Kulkarni; Dr. Ganesh M. Kakandikar	202521014218	Published	India

4	An Assistive Robotic System	Rajesh Vinayak Patil; Hamza Anwar Tamboowala; Vedang Umesh Pasalkar	202521032745	Published	India
5	A System for Protecting a Rider and Pillion of a Two-Wheeler	Nilesh K. Birajdar	202521034900	Published	India
6	A System for Dispensing Items	Manas Jain; Varad Charholikar; Dr. Ashish Pawar; Dr. Madhuri Bhalekar	202521037762	Published	India

7	A Modular Unmanned Ground Vehicle with Multi-Modal Sensing and Control Capabilities	Jaideep Vinod Aher; Abhijit Suresh Gadekar; Shivprakash B. Barve; Vibha Patel	202424065762	Published	India
8	A Bamboo Moisture Chamber	Razdan Surbhi; Karkale Pranay Pramod; Bhate Hardik; Aaher Shruti; Aagre Rugved; Bansal Gaurav; Pokale Atharva	202424092938	Published	India
9	A Machine for Washing Fresh Produce	Neel Tamhankar; Hitik Patel; Dr. Kaustubh Ganesh Kulkarni; Dr. Atul Marotrao Elgandelwar; Soumyajit Ghosh; Ayush Abhay Wadgaonkar	202521043907	Published	India
10	A Composite Layer for Effective Noise Reduction in Absorptive Muffler	Dr. Mahesh Vasantrao Kulkarni; Dr. Deepak Popat Hujare; Dr. Girish Sharad Barpande	202521049068	Published	India
11	System for Dispensing Multiple Liquids	Dr. Anil Shivsharan Mashalkar; Deep Shirode; Ritvik Navnath Gosavi; Shravan Maheshwari; Akshay Rane	202521053728	Published	India
12	GPS Tracking Device with Dual Charging and Multi-Point Mounting	Dr. Kulvaibhav Sharma; Dr. Vijendra Kumar; Dr. Mitulkumar Thakorbbhai Solanki	449862-001	Published	India

Sr. No.	Patent Title	Inventor(s)	Application No.	Status	Country	Published On
1	Concrete Climate Controller	Rajat Kshirsagar; Sunil Sudhakar Pimplikar	455160-001	Published	India	Apr 12, 2025
2	GPS Tracking Device with Dual Charging and Multi-Point Mounting	Kul Vaibhav Sharma; Vijendra Kumar; Mitulkumar Solanki	449862-001	Published	India	Jun 30, 2025

Sr. No.	Patent Title	Inventor(s)	Application No.	Status	Country	Published On
1	Method for production of biohydrogen by anaerobic fermentation of sugar containing syrup	Bharat Kale; Sagar P. Kanekar; Anand D. Kulkarni; Nirav Dobariya	20252106365 1 A	Published	India	Jul 25, 2025
2	A method of characterization of bacteriorhodopsin in haloarchaeal organism	Sagar Kanekar; Anand D. Kulkarni; Pratiksha D. Khurpade; Vivek Chakraborty; Ruchi Mathad	20252101152 6 A	Published	India	Apr 18, 2025



The following Alumni visited the Department.

Sr. No.	Name of the Alumni	Passing year
1	Arpit B. Bendle	2011
2	Mihir Adivarekar	2010
3	Akshay Kumbhar	2010
4	Anurag Parekh	2021
5	Chinmay Lokhande	2023

Achievement & Milestones

Students from the Polytechnic department achieved remarkable milestones across various competitions

SY and TY students of CSE/AIDS—Sujal Sokande, Nidhi Thakare, Vivaan Mathur, and Pratik Mishra—secured 1st place in the Ideathon at the MITWPU Hackathon 2025, held at MITWPU, Pune



Ram Kansal, a TY AIDS student, secured 1st place in the Workathon at the MITWPU Hackathon 2025, held at MITWPU, Pune.



Aarushi Javale, Aditya Shevate, Janhavi Honrao, and Aditya Patil from SY CSE received the Best Startup Award at the Pune Startup Expo 2025, held at MITWPU, Pune, in the presence of the Minister of Information and Technology Hon. Ashish Shelar Sir and Dr. Rahul V. Karad Sir.





The geological visit organized for 2nd year Btech (Petroleum engineering) students accompanied by Dr Rahul Joshi and Dr. Kunal Modak

National Academic Immersion Program (NAIP) was arranged at IIT BHU, Varanasi, UP. 70 students learnt theory and hands on training from IIT professors. They were enriched with knowledge and also visited SATHI to gain insight on different high-end characterization equipments. Students also visited Kashi Vishwanath Temple, Prayagraj and Ayodhya.



- Sai Telkar,Aadya Tiwari,Gauri Jaju SY & TY ECE-AIML students, secured 1st place in the Workathon at the MITWPU Hackathon 2025, held at MITWPU, Pune.



- Shreya Karandikar,Anushka Jagtap,Devish Darne,Vaishnav Shitut SY &TY ECE-AIML students won Appreciation Prize in ideathon at the MITWPU Hackathon 2025 held at MITWPU, Pune.
- Aryan Mahajan, Dhruv Abhyankar TYECE students secured runner up position and won cash prize of Rs.5000/- in Project competition organized by Polymer Department of MITWPU.
- Yash Mundada, Rupesh Itkar, Aryan Thopte, Shrey Mathwad,TYCE (Int.Btech MIT WPU) students won runner up prize at Workathon at the MITWPU Hackathon 2025, held at MITWPU, Pune
- Yash Mundada, Rupesh Itkar, Aryan Thopte, Shrey MathwadTYCE (Int.Btech MIT WPU) students won first prize at project competition Protech 2025 organized by Symbiosis Institute of Technology,Pune.





The 1st PhD of Petroleum Engineering department awarded to Mr Soumitra Nande under the supervision of Prof Samarth Patwardhan

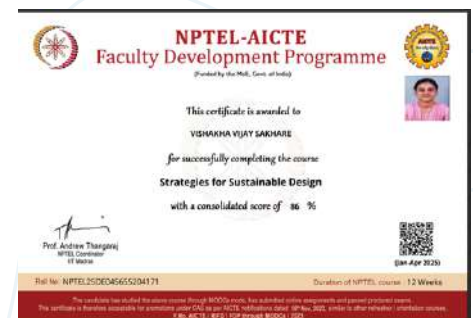
The faculty members of Civil Engineering attended the Faculty Development Programme (FDP) on “OBE in Action: Outcome Attainment and Teaching-Learning Insights” held on 25 and 26 July 2025. The Programme focused on outcome-based education practices and effective teaching-learning strategies.



Dr. Ganesh Ingle participated in the Faculty Development Programme on “Emerging Developments in Engineering Education.” The programme enhanced understanding of recent trends and effective practices in engineering education.



This is to certify that Dr. Vishakha Sakhare successfully completed the NPTEL–AICTE Faculty Development Programme course on “Strategies for Sustainable Design.” The course was completed with a consolidated score of 86% during the 12-week Programme (Jan–Apr 2025).



Placements & Career Progression

Sr. No.	Name of the placed student	Designation	Name of the Company	CTC	Photograph of a student	Company logo
1	Jha Rohan	Business Technology Solutions Associate	ZS Associates	13.68		
2	Deshmukh Sanya Yashodhan	Business Technology Solutions Associate	ZS Associates	13.68		
3	Ashish Kumar Tiwari	Business Technology Solutions Associate	ZS Associates	13.68		
4	Bishoi Ashutosh	Graduate Engineer/Analyst	ANZ	12.5		
5	Somani Varun Rakesh	Software Engineer	UBS	12.5		
6	Bishnoi Isha	Software Developer	IBM	12.25		
7	Md Faiz Alam	Software Developer	Amadeus 2025	11.5		
8	Chintamani Shivam Bipin	Software Developer	Amadeus 2025	11.5		
9	Gupta Aryan	Software Developer	A. P . Moller	10.83		
10	Pande Awadhoot Ravindra	Software Developer	ATS conveyor	10.6		
11	Sneha Sunil Gadgil	Software Developer	ATS conveyor	10.6		

12	Saksham Jain	Jr. Software Developer	Onextel	10		
13	Siddharth Kumar Sharma	Jr. Software Developer	Onextel	10		
14	Anshika Mishra	Jr. Software Developer	Onextel	10		
15	<i>Mahammed Taha Shaikh</i>		<i>Ashoka Buildcon Pvt. Ltd.</i>	4		

Sr. No.	Name of the placed student	Designation	Name of the Company	CTC (In Lakhs)	Company logo
1	Kaushal Sachin Kshirsagar	Graduate Engineer Trainee	ATS Conveyors	11	 ATS Group
2	Deshmukh Sanya Yashodhan	Graduate Engineer Trainee	ATS Conveyors	11	 ATS Group
3	Shreerang Sandeep Wakade	Graduate Engineer Trainee	ATS Conveyors	11	 ATS Group
4	Aryan Rahul Dere	Graduate Engineer Trainee	3D Magic	4.2	 3D magic
5	Vanshita Vaibhav Bafna	Graduate Engineer Trainee	3D Magic	4.2	 3D magic
6	Dipanshu Dilip Nanhe	Graduate Engineer Trainee	Sulzer	8	 SULZER
7	Manan Nehru	Graduate Engineer Trainee	Sulzer	8	 SULZER
8	Pranav Umesh Shirsavkar	Graduate Engineer Trainee	Sulzer	8	 SULZER
9	Vedant Pravin Harti	Graduate Engineer Trainee	Sulzer	8	 SULZER

SR. No.	Name of the placed student	Designation	Name of Company	CTC	Photograph of student	Company logo
1	Sumant Venkangouda Jadyappagouda r	Junior Trainee	Lupin Biopharmaceut icals	3.78 LPA		
2	Harsh Chandrakant Veer	Associate Cloud Support Engineer.	Strategic India	6.6 LPA		
3	Tanmay Bhongale	Field Service Engineer	Aerobio Labs	3.5LPA		
4	Harsh Salagare	Field Service Engineer	Aerobio Labs	3.5 LPA		
5	Shivansh Sharma	Junior Trainee	Iziel Healthcare	4 LPA		
6	Aastha Dastane	Junior Trainee	Iziel Healthcare	4 LPA		
7	Tejaswini Singh	Project Lead intern role	Iziel Healthcare	1.2 LPA		
8	Aniket Arun Sarnaik	Junior Trainee	Gyro Drive	10 LPA		
9	Sahil vijay Thakare	Inside Sales Specialist Intern	Jyesta Entity	9.5 LPA		

Pre-Placement Bootcamp was organized by Dr. Shraddha Kulkarni with the wholehearted support of the CDC team. The expert sessions conducted by distinguished industry professionals and their insightful inputs will undoubtedly play a pivotal role in enhancing our students' preparedness and confidence as they step into the professional world. Huge thanks to Dr. Bhalchandra Vaidya, Gennova Biopharmaceuticals Ltd, Pranav Kshirsagar, Agharkar Research Institute, Pune, Dipali Bagal-Kestwal, Mylab Discovery Solutions, Manasi Kulkarni-Khasnis, Biomark IQ, Vikas Shirolkar and Mr. Manas Vhora, Devise Electronics Pvt. Ltd. for their valuable contributions and engagement with our students during the Bootcamp. Also grateful to Meraj Nimbarg, Vikas Waghmare, and Dr. Ruta Barbadekar for their dedicated support in delivering impactful soft skills training sessions. Thanks to Bioengineering Faculty Team for providing the support.

A special note of thanks to reporting authorities Dr. Kishanprasad Gunale, Dr Dinesh Bhutada, Ganesh Kakandikar, Dr. Vikrant Gaikwad, Manas Mohanty, Dr Sidharth Chakrabarti, Career Development Centre (Student's Career Council MIT-WPU), MIT World Peace University for continuously trusting with opportunities and offering a platform to lead initiatives that support the placement journey of our Bioengineering students.



Campus Events & Activities

National Academic Immersion Programme (NAIP)

The Department of Electrical and Electronics Engineering, Dr. Vishwanath Karad MIT World Peace University, successfully organized the National Academic Immersion Programme (NAIP) 2025–26 at the International Automobile Centre of Excellence (iACE), Gandhinagar, for Third-Year B.Tech students from E&CE, ECE, and ECE (AI-ML).

The programme was conducted in three batches between 5th June and 10th July 2025, engaging 385 students in an intensive 10-day experiential learning journey focused on modern automotive technologies and Industry 4.0 practices.

iACE, a premier automotive training hub established by the Government of Gujarat and Maruti Suzuki India Ltd., provided students with hands-on exposure to advanced automotive technologies. The immersive programme covered key domains such as automotive systems, AutoTronics, digital electronics, CAN/LIN vehicular communication, ADAS, image processing, robotics, embedded systems, sensorics, and Industry 4.0.

Students gained practical experience through live demonstrations on Toyota Fortuner and Prius platforms, ADAS calibration systems, industrial robots, and microcontroller-based embedded kits.

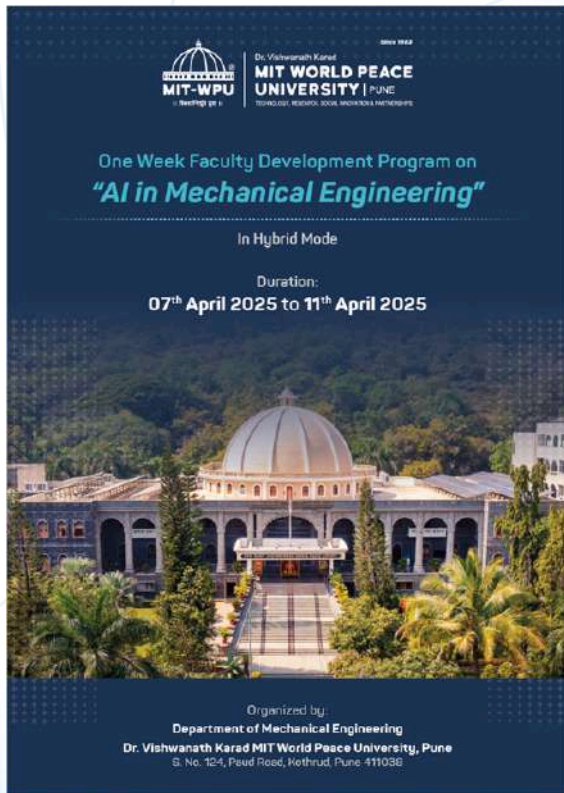
The programme also featured industrial visits to Ammann India Pvt. Ltd. and Uno Minda Limited, offering insights into modern manufacturing, automation, and electric vehicle technologies.

A cultural tour of Ahmedabad, including visits to Gandhi Ashram and Adalaj Stepwell, enriched the learning experience by blending technical exposure with cultural understanding.

Overall, NAIP 2025–26 significantly enhanced students' technical competence, industry awareness, and professional skills, with several participants selecting e-mobility-focused academic projects, reflecting the programme's strong industry alignment and impact.



FDP on AI in Mechanical Engineering from 7th to 11th April 2025.

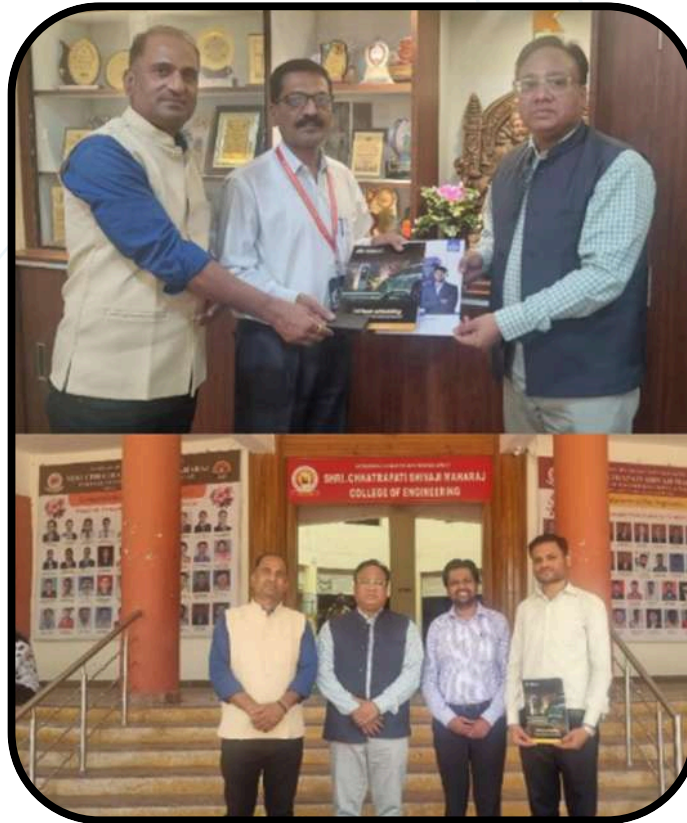


FDP organized on Robot Operating System (ROS) from 5th May to 9th May 2025.



Guest Lecture on "Career Opportunities in the Field of HVACR"

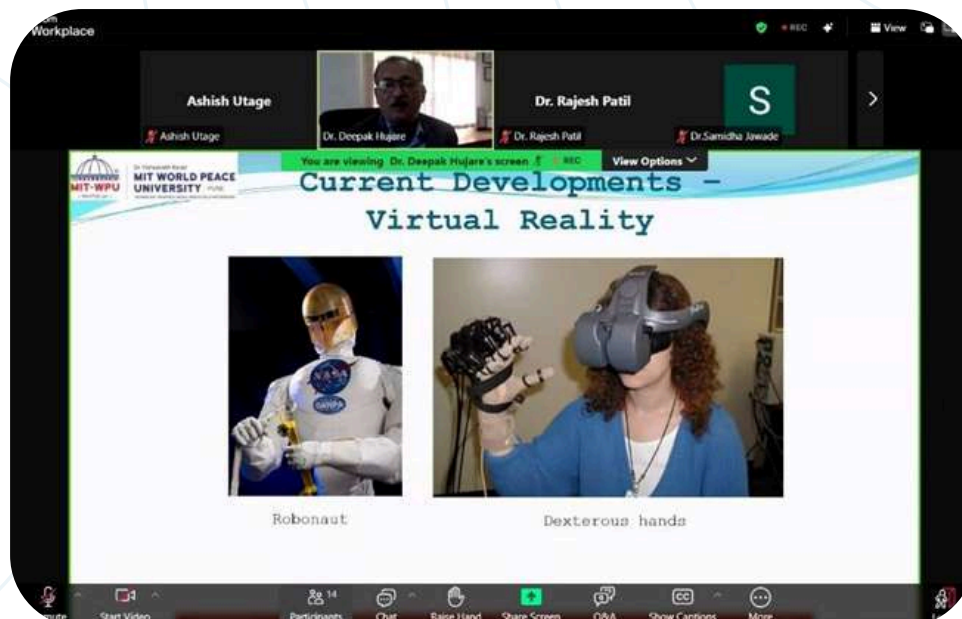




Department of Mechanical Engineering (DoME) conducted a highly engaging session titled “AI in Robotics and Drone Building”, in which students were introduced to the integration of artificial intelligence into mechanical systems. Participants engaged in building mobile robots, ROVE’s and basic autonomous functions.



Online webinars conducted throughout the year with targeted themes:
 Audience: Students from Maharashtra and across India preparing for JEE, CET, and GATE
 Outcome: Increased student engagement, personalized counselling, and higher digital reach.



The Civil Engineering Department, MIT-WPU successfully organized Nirmitee 2025, a prestigious National Level Technical Event, marking its 25th anniversary on 28th & 29th March 2025. The event provided a vibrant platform for students to showcase technical skills, innovation, and research in civil engineering. The Valedictory Function, held on 29th March 2025 at Swami Vivekananda Auditorium, was graced by Mr. J. S. Salunkhe, Regional Head, Maharashtra Pollution Control Board, as the Chief Guest. The event emphasized themes of sustainability, innovation, and professional excellence, inspiring budding civil engineers.



Faculty members of the Civil Engineering Department actively participated in the Power of Synergy team-building program, organized for the faculty of Engineering & Technology. The event enhanced collaboration, communication, and problem-solving skills through interactive activities, reinforcing the importance of teamwork and individual strengths in achieving collective academic and institutional goals.

Faculty also actively attended the FDP on Pedagogical Approaches and Catering to Diversity, gaining valuable insights into effective teaching-learning strategies. The workshop enhanced understanding of inclusive pedagogy and outcome-based education, supporting academic excellence and alignment with NAAC learning and teaching standards.

Congratulations
ON YOUR PLACEMENT (B.TECH BIOENGINEERING, 2025)

 Mr. Sumant J Lupin Biopharmaceutical	 Ms. Riya Tyagi GRG Healthcare	 Mr. Sahil Thakare JYESTA CORPORATE ENTITY	 Ms. Sakshi Kinago JYESTA CORPORATE ENTITY And IZiel Healthcare	 Ms. Iravati Dhembare IZiel Healthcare	 Ms. Swarali Nikam IZiel Healthcare
 Ms. Aastha Dastane IZiel Healthcare	 Ms. Tojaswini Sing IZiel Healthcare	 Mr. Aniket Sarnaik Gyro Drive	 Mr. Shivansh Sharma IZiel Healthcare	 Ms. Shriya Pote IZiel Healthcare	

“Our Esteemed Recruiters”









Student & Faculty Corner

Shreya Aigalika, student of Department of Electrical and Electronics Engineering, MITWPU won the first prize for the project titled "Raag Vision - Decoding Hindusthani Classical Music with Machine Learning" under poster competition at National Scientist Round Table Conference.

Mini project Model Exhibition of Mini Project Students in the Department of Mechanical Engineering on 5th May 2025.

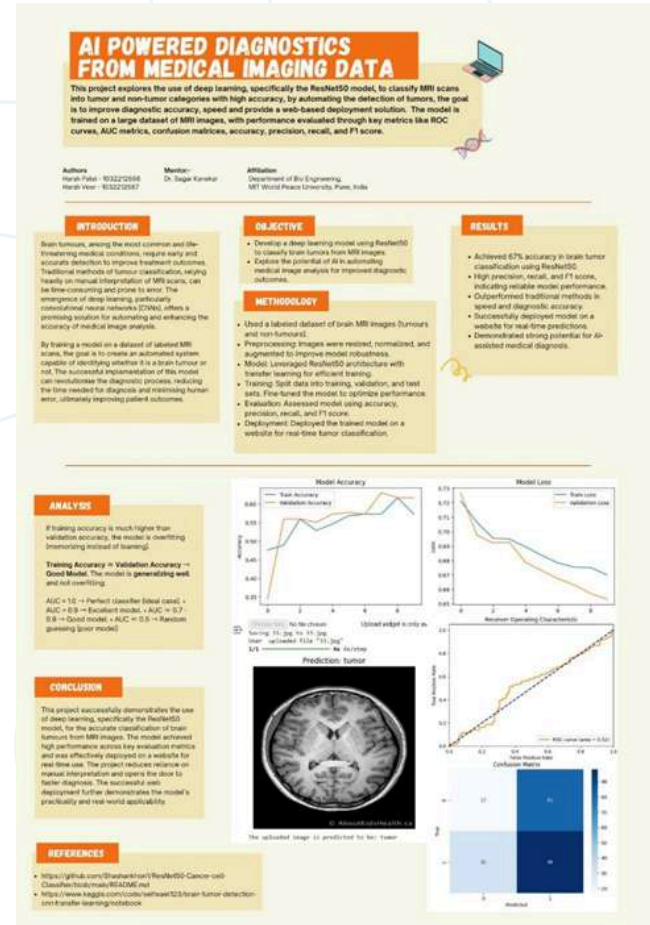
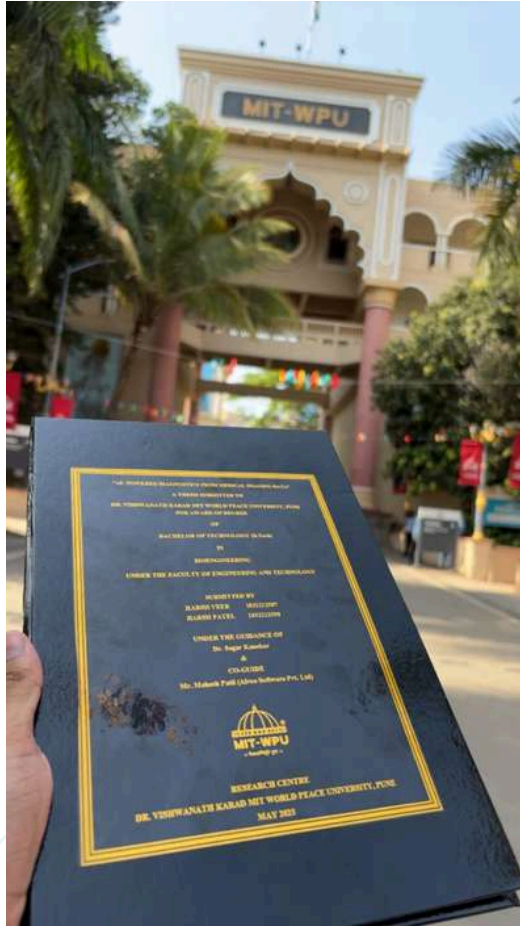


Arya Shingade T Y Bioengineering student successfully completed the "Whodunnit? DNA Fingerprinting Workshop" at IISER Pune on 27th April 2025. This workshop was an incredible opportunity to explore the world of forensic science and DNA analysis, giving me hands-on experience in techniques that have far-reaching applications in criminal investigations.



Golden year batch 2021-2025 convocation ceremony was organized and students presented their capstone project work in the department.





Capstone Project

Clubs & Extracurricular

Triangle Trophy 2025: Celebrating Talent, Teamwork, and Holistic Development

The Department of Electrical and Electronics Engineering (DoEEE) proudly organized the Triangle Trophy 2025, a flagship student welfare initiative designed to promote holistic development through a balanced blend of academic, cultural, and sports activities. The event witnessed enthusiastic participation from students across three houses — Red, Green, and Blue — fostering healthy competition, camaraderie, and all-round growth.

The Triangle Trophy is structured around three vibrant student clubs: MASS (Meta Association of Study Skills) focusing on academic enrichment, MARS (Meta Association of Recreational Sports) promoting physical fitness and teamwork, and CHARM (Cultural Hive and Recreational Mesh) nurturing creativity and artistic expression.

Under MASS, students showcased their analytical and creative abilities through events such as Aptitude Test, Debate Competition, Question Quest, Component Conclave, Circuit Heist, and Creative Writing. These competitions encouraged critical thinking, problem-solving, technical recall, and effective communication, reinforcing the academic foundation of engineering education.

The department also recognized meritorious students through the Academic Performance Appreciation Scheme (APAS) and Consistently High Academic Performance (CHAP), applauding both exceptional achievements and sustained academic excellence.

APAS & CHAP



Students receiving certificates for APAS & CHAP



Students receiving certificates for APAS & CHAP

MARS organized a series of high-energy sports events including Chess, Badminton, Cricket, Football, and Basketball. Conducted in league-cum-final formats, these competitions tested endurance, strategy, and teamwork, while instilling discipline and sportsmanship among participants. The events drew large student participation and created an atmosphere of excitement and unity across the department.

The cultural segment led by CHARM featured Fine Arts, Band Performances, Solo Singing, Solo Dancing, and Group Dance events. With the theme "Nature's Symphony" for Fine Arts and vibrant stage performances during the cultural finale, students displayed exceptional creativity, confidence, and artistic flair, captivating audiences and judges alike.

After weeks of spirited competition, Blue House emerged as the Overall Winner of Triangle Trophy 2025, with Red House securing the Runner-Up position. The success of the event was made possible through the dedicated efforts of faculty mentors, student coordinators, and the EESA team, whose leadership and coordination ensured smooth execution.

Triangle Trophy 2025 stood as a testament to DoEEE's commitment to student welfare, experiential learning, and holistic education. Beyond competition, the event nurtured leadership, collaboration, creativity, and resilience—core attributes essential for future-ready engineers.



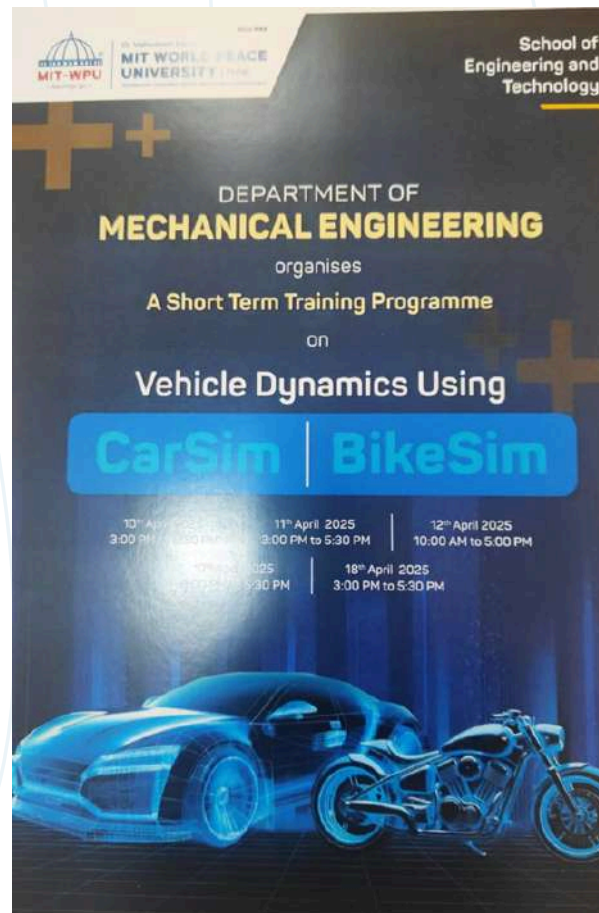
Winners of the triangle trophy (Blue House)

Runner ups of Triangle Trophy



Runner ups of the triangle trophy (Red House)

A five days short term training program for students from various Clubs under DoME and Faculty members has been planned on a theme Vehicle Dynamics using CarSim and BikeSim from April 10 to April 18, 2025.



MIT WPU SBE student chapter orientation and felicitation was organized by SBE and all the selected panel members were felicitated.



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तृण भारत

ED

१५ अक्टूबर २०२४

तर्ज

एक ही छवि में दो अलग-अलग प्रकार के चित्रों को जोड़ने के लिए एक ही छवि को दो अलग-अलग छवियों में बांट दिया गया है।

— विष्णु शिव शर्मा



पेट्रोलियम अभियान्त्रिकी

आधुनात्मक

अभियान्त्रिकी शाखा



अधुनात्मक अभियान्त्रिकी शाखा

यह शाखा अभियान्त्रिकी शाखा के अन्तर्गत आती है। यह शाखा अभियान्त्रिकी शाखा के अन्तर्गत आती है। यह शाखा अभियान्त्रिकी शाखा के अन्तर्गत आती है।

पेट्रोलियम अभियान्त्रिकी शाखा

यह शाखा अभियान्त्रिकी शाखा के अन्तर्गत आती है। यह शाखा अभियान्त्रिकी शाखा के अन्तर्गत आती है। यह शाखा अभियान्त्रिकी शाखा के अन्तर्गत आती है।

अभियान्त्रिकी शाखा

यह शाखा अभियान्त्रिकी शाखा के अन्तर्गत आती है। यह शाखा अभियान्त्रिकी शाखा के अन्तर्गत आती है। यह शाखा अभियान्त्रिकी शाखा के अन्तर्गत आती है।

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BW BUSINESSWORLD
Tuesday, Jun 24 2023 11:08:07 PM

India Eyes Rs 3 Lakh Cr Tunnelling Push: Nitin Gadkari



India is gearing up for a massive expansion in tunnel infrastructure, with projects worth Rs 2.5 to Rs 3 lakh crore planned over the next decade, Union Minister for Road Transport and Highways Nitin Gadkari said on Monday. Speaking at the International Workshop on Sustainable Tunnelling for Better Life (STBL) at MIT World Peace University (MIT-WPU), Gadkari called for a renewed focus on reducing construction costs, upgrading technology, and outperforming tunnelling machinery.

The minister's remarks came during the inauguration of India's first centre of excellence for tunnelling and underground construction at MIT-WPU. Developed in partnership with Tata Projects and Sandvik, the centre features a Tunnel Monitoring Laboratory and a Drilling & Blasting Laboratory aimed at bolstering domestic capabilities in underground infrastructure.

"India is entering a golden era of infrastructure. Tunnels are essential for connectivity, safety, and sustainability," Gadkari said. "We must adopt fuels like ethanol, CNG, hydrogen, and electric, and manufacture or import used tunnel boring machines from Europe to cut costs." The two-day workshop, co-organised with the International Tunnelling and Underground Space Association's Committee on Education and Training (ITACET), spotlighted the importance of sustainable practices and advanced engineering in tunnelling, especially in sensitive regions like the Himalayas.

Experts from India, Europe, the UK, and the US deliberated on digital innovations such as Building Information Modelling (BIM), laser scanning, and low-impact excavation techniques. Emphasis was laid on reducing the environmental footprint of tunnels and increasing the safety of underground construction workers.

Arnold Ditz, Past President, International Tunnelling Association, underlined the global relevance of India's initiative. "This Centre bridges a critical skills gap. Too often, young workers are at risk because they lack practical training. What I saw here at MIT-WPU—the mentorship and family-like academic environment—is exceptional," he said.

India's tunnel projects—ranging from highway expansions to strategic defence and urban mobility—are being fast-tracked to support economic growth, energy security, and disaster resilience. The government is pushing for deeper involvement from private players, global technology providers, and academic institutions to reduce imports and enhance local skill development.

Gadkari assured that his ministry would back initiatives with equipment and training support. "With innovation, commitment, and collaboration, we can make India self-reliant in tunnelling technology," he said.

bright Punjab Express

Tunnel projects worth ₹3 lakh cr to be built in next 10 years as part of big infra push: Gadkari

PUNE, JUNE 24

Union Minister for Road Transport and Highways Nitin Gadkari on Tuesday said that the government has drawn up plans to build tunnel projects worth Rs 2.5 to Rs 3 lakh crore over the next 10 years as part of the country's infrastructure development.

Speaking at the inauguration of the International Workshop on 'Sustainable Tunnelling for Better Life' at MIT World Peace University (MIT-WPU) here, the minister said, "India is entering a golden era of infrastructure development, with tunnels playing a crucial role in connectivity, safety, and sustainability."

The minister highlighted that there was a need to reduce construction costs without compromising quality. "That means using new technologies and sustainable fuels like CNG, ethanol, hydrogen, and electric alternatives to diesel. We should also refurbish old tunnelling machines, import used ones



from European countries like Austria, Norway, and Spain, and eventually manufacture our own," he explained.

He pointed out that India's geology varies by region, so research and training are essential. Industry experts and experienced engineers should guide students along with faculty.

"My ministry is ready to support with equipment and training. Together, with innovation, research, and commitment, we can make India self-reliant in tunnelling technology and infrastructure development," Gadkari said.

He also lauded MIT-WPU for taking the first step towards research in sustainable tunnelling technology, which is the need of the hour for a developing

country like India.

In a bid to boost sustainable infrastructure development, Gadkari, on Tuesday, inaugurated the International Workshop on 'Sustainable Tunnelling for Better Life' at MIT World Peace University (MIT-WPU). The two-day event was organised in collaboration with the International Tunnelling and Underground Space Association's Committee on Education and Training (ITA-CET). The two-day workshop has brought together global experts from India, Europe, the UK, and the US.

A key highlight of the event was the inauguration of the Centre of Excellence for Tunnelling and Underground Construction at MIT-WPU, India's first-of-its-kind facility featuring a Tunnel Monitoring Laboratory and a Drilling and Blasting Laboratory.

The Centre of Excellence, set up in collaboration with Sandvik and Tata Projects Ltd, aims to support advanced research and training in underground construction technologies. **IAN S**

"Nitin Gadkari unveils an ambitious vision for India's infrastructure growth, announcing tunnel projects worth ₹3 lakh crore over the next decade to boost connectivity and sustainable development."

The Indian EXPRESS
JOURNALISM OF COURAGE

Tunnel projects worth Rs 3 lakh cr in next 10 years in India: Gadkari

SOHAM SHAH
PUNE, JUNE 23

UNION MINISTER of Road Transport and Highways Nitin Gadkari said that tunnelling projects worth Rs 2.5 to Rs 3 lakh crore would be built in India over the next 10 years. Speaking at the inauguration of the International Workshop on Sustainable Tunnelling for Better Life at MIT World Peace University in Pune, Gadkari said, "There is huge potential in tunnels. Over the next 10 years, we plan to build tunnel projects worth Rs 2.5 to Rs 3 lakh crore. In Delhi, under the Talkatora stadium to Gurgaon, we have started the construction. In Bangalore, we have started making a tunnel from both sides of the 'V'." He added, "Many tunnelling machines come from China.



Padma Vibhushan Dr. Vijay Kelkar receives the Punyabharat Parashad from Union Minister Nitin Gadkari at Balgandharva Rangamandir in Pune on Monday. *Arul Hosain*

make this a reality, we must reduce construction costs without compromising quality. That

old tunnelling machines, import used ones from European countries like Austria, Norway, and

Tunnel Monitoring Laboratory and a Drilling & Blasting Laboratory. The Centre of

of the International Tunnelling Association, highlighted the importance of having practical experience before graduates are sent out into the workforce of tunnelling. "For decades I have attended disasters and I know young men and women who lose their lives because they just don't know how to build the tunnels. And they work so hard but the skills section is to be absolutely supported. It is a wonderful thing. Too often, young workers are placed at risk because they lack the training needed to safely construct what has been so carefully designed."

SK Dharmadhikari, an infrastructure sector veteran, was conferred with the Lifetime Achievement Award for his invaluable contribution in nurturing and shaping the next generation of tunnel engineers. Companies like TATA Projects

Educational TIMES
ONLINE

World Earth Day: Students turn sugarcane into green hydrogen, farm waste into clean fuel

The sustainable method, developed at MIT WPU aligns with India's Green Hydrogen Mission



Students, along with researchers and faculty members, have developed a novel process for generating green hydrogen from sugarcane juice and an efficient biodiesel production system utilizing agro-waste-based catalysts—both offering eco-friendly alternatives to conventional fossil fuels.

The MIT World Peace University has made significant strides in sustainable energy with these innovations. The innovative process generates hydrogen from sugarcane juice while simultaneously converting carbon dioxide emissions into a sustainable resource. The sugarcane juice, developed at MIT, is used to generate hydrogen, while the waste water, along with the water from sugarcane juice, is used to generate hydrogen. The innovation not only reduces dependence on fossil fuels but also contributes to efforts aimed at bringing down the cost of hydrogen production to \$1 per kilogram (kg). "The process operates without the need for energy-intensive water splitting, and produces valuable byproducts ensuring zero discharge. We are seeking industry partners for lab-scale development and eventual technology transfer," says Prof. Bharat Kale, Emeritus Scientist and Director of the Centre of Excellence, Material Science, MIT WPU.

The hydrogen project, led by a team including Sage Randerkar, Bharat Kale, Anand Kulkarni, Prof. Nand Kulkarni, Suman Shah, Devendra, Anurag, and Dhanraj, jointly has already submitted a patent and is proposed for funding through the Ministry of New and Renewable Energy (MNRE). A Centre of Excellence on Green Hydrogen at the university has also been proposed to the MNRE. In addition to this, MIT WPU researchers have also introduced a patented batch reactor system for biodiesel production. The system employs a heterogeneous catalyst derived from agricultural waste, offering an environmentally friendly and cost-effective alternative to traditional methods. Its porous structure enhances surface area, boosts efficiency, and ensures thermal stability during production. "Biodiesel produced through our system presents a viable alternative to fossil fuels and reduces environmental impact. The patent encompasses both the process and the catalyst design. Commercial release is anticipated within six months of technology transfer pending ongoing industry partnership discussions," says Prof. Kale.

The biodiesel initiative is led by Prof. Nand Kulkarni, Suman Shah, and Bharat Kale and it aims to provide a sustainable solution to biomass waste, especially in states like Punjab and Haryana, where stubble burning is a persistent environmental concern.

These pioneering developments underscore MIT WPU's commitment to supporting India's green energy goals by offering scalable, commercially viable solutions. The university continues to seek collaborative industry partnerships to fast-track the commercialization of these technologies and contribute to a cleaner, greener future.

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MITWPU Explores Green Hydrogen, Biodiesel Innovations

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Researchers at MIT World Peace University (MITWPU) have developed two new methods aimed at supporting India's green energy goals: one involving the production of green hydrogen from sugarcane juice and another using agro-waste-based heterogeneous catalysis in a batch reactor system for biodiesel generation.

The projects contribute to ongoing efforts within academic and scientific communities to explore alternative, low-emission fuel technologies.

Bharat Kale, Emeritus Professor and Director, Material Science COE, highlighted, "Unlike conventional water splitting methods, this process generates valuable byproducts, ensuring zero discharge and making it a viable solution for India's energy transition. We are seeking industry partners for lab-scale development and eventual technology transfer."

A patent application has been filed for the green hydrogen technology, while a proposal for project funding has been submitted to the Ministry of New and Renewable Energy (MNRE). A separate proposal to establish a Centre of Excellence on Green Hydrogen at MITWPU has also been submitted to the MNRE, reflecting the university's bid to contribute to national level research infrastructure in the sector.

From farm waste to clean energy—MIT-WPU leads the way in green hydrogen and biodiesel innovations, contributing to a sustainable and low-emission future.



Dr Siraj Bhatkar from Department of Petroleum

Department of Bio Engineering

Kavach Khanna, a YouTube influencer made a video of Official MIT-WPU Campus Tour. This video provides all the details which the students need to know about the campus of MIT-WPU, Pune, the facilities the campus provides, classrooms, seminar rooms, various rules/regulations, sports facilities, the cafeteria, various labs, etc. It also showed the inside of MIT-WPU, Pune Campus and projects students working on in Bioengineering department especially biohydrogen project. Link of the video.

<https://youtube.com/watch?v=ATMGRvqPqyQ&si=t3OvTzLe294RMLh8>

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